TEXORA · LECTURE SHEET

Nanotechnology

Working at the scale of billionths of a metre changes what a fibre surface can do. This sheet defines the nanoscale, explains the nanomaterials used in textiles and the functions they deliver, and treats durability and safety honestly.

YM-4203 · TECHNICAL FIBRE & YARN · TOPIC 2 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. The nanoscale

One **nanometre** (nm) is one billionth of a metre, 10^{-9} m. **Nanotechnology** is the design and use of structures with at least one dimension roughly in the range 1–100 nm. For scale: a human hair is about 50,000–100,000 nm wide, a typical textile fibre is 10,000–30,000 nm in diameter, and a nanofibre is usually defined as a fibre below 1,000 nm, often 100–500 nm, in diameter.

Two effects make the nanoscale interesting, and neither is magic:

- **Surface-area effect.** As particles or fibres get smaller, a far larger fraction of their atoms sits at the surface, where reactions happen. Surface area per unit volume rises in inverse proportion to size (Section 4), so a gram of nanoparticles presents an enormous active surface.
- **Size-dependent behaviour.** At very small sizes some properties change with size itself — for example, semiconductor particles such as TiO_2 and ZnO absorb ultraviolet light very effectively, and silver becomes strongly antimicrobial because it releases silver ions from its large surface.

> **DISCIPLINE OF THE SUBJECT**
> “Nano” on a label is not a function. Every claim in this field must name the material, the size, how it is attached to the textile, and the test that proves the function — before and after washing.

2. Nanomaterials used in textiles

2.1 Nanofibres

Continuous fibres of sub-micrometre diameter, produced mainly by electrospinning (Topic 3). Their webs have very small pores, huge surface area and low weight per unit area. Uses: high-efficiency air and liquid filtration, wound dressings, protective clothing layers and tissue-engineering scaffolds.

2.2 Nanoparticles

- **Silver (Ag)** — antimicrobial. Silver ions interfere with bacterial cell membranes and enzymes. Used in anti-odour socks, sportswear, wound dressings and hospital textiles.
- **Titanium dioxide (TiO_2)** — ultraviolet absorption and photocatalysis. Under UV light, photoexcited TiO_2 generates reactive species that can break down organic stains and odorous compounds — the basis of "self-cleaning" finishes.
- **Zinc oxide (ZnO)** — UV protection and antimicrobial action, often preferred where a less photocatalytically aggressive particle is wanted.
- **Others** — silica (SiO_2) for modifying surface roughness and water repellency, nanoclays for barrier and flame-retardant effects, and carbon nanotubes or graphene-related materials for electrical conductivity and strength in research and specialist products.

2.3 Nanocomposites

A **nanocomposite** disperses a nanoscale filler inside a bulk material — for example nanoclay platelets or nanoparticles inside a fibre polymer or a coating resin. At only a few percent loading, a well-dispersed nanofiller can raise modulus, improve barrier properties and slow flaming, because the filler–polymer interface area is so large. Poor dispersion destroys the benefit: agglomerated nanoparticles behave like ordinary coarse filler with added cost.

FUNCTION	TYPICAL NANOMATERIAL	MECHANISM	TYPICAL PRODUCTS
Antimicrobial / anti-odour	Ag, ZnO	Ion release; membrane and enzyme disruption	Socks, sportswear, dressings
UV protection	TiO_2, ZnO	Absorption and scattering of UV	Summer clothing, awnings
Self-cleaning	TiO_2 (photocatalytic)	UV-driven oxidation of organic soil	Outdoor fabrics, workwear trials
Conductivity / antistatic	Carbon nanotubes, graphene, Ag	Conductive network at low loading	Smart textiles, antistatic clothing
Barrier / flame retardancy	Nanoclay	Tortuous diffusion path; char promotion	Coatings, upholstery
Filtration	Electrospun nanofibres	Very small pores, high surface area	Air and liquid filters

Table 1 — Functions delivered by textile nanomaterials. Each function must be verified by its own standard test on the finished, washed article.

3. Application routes

3.1 In-fibre incorporation

Nanoparticles are mixed into the polymer melt or spinning solution before extrusion, so they are embedded throughout the fibre cross-section (Figure 1, left). Advantages: the function survives abrasion and washing better, because new surface exposed by wear still contains particles. Difficulties: dispersion in the melt, spinneret blockage, particle effects on drawing and strength, and the fact that particles buried deep inside the fibre contribute nothing to a surface function — an expensive waste for antimicrobial use.

3.2 Coating and finishing

Nanoparticles are applied to a finished fabric by padding, coating or sol–gel processing, held by a binder or by chemical bonding to the fibre surface (Figure 1, right). Advantages: any fabric can be treated, the active material sits exactly where it works — at the surface — and loadings are low. Weakness: durability. A physically held particle can be removed by washing and abrasion; binders, crosslinking and plasma pre-treatment are used to improve fixation, and wash durability must be tested, not assumed.

3.3 Plasma-assisted routes

Cold plasma — an ionised gas at near-room temperature — can etch, activate or deposit on a fibre surface at nanometre scale without wet chemistry. Plasma activation adds polar groups that improve the bonding of a subsequent nano-finish; plasma deposition can build very thin functional layers. Plasma is a dry, low-water route, but treatment is line-of-sight, equipment cost is significant, and uniformity on bulky goods needs care.

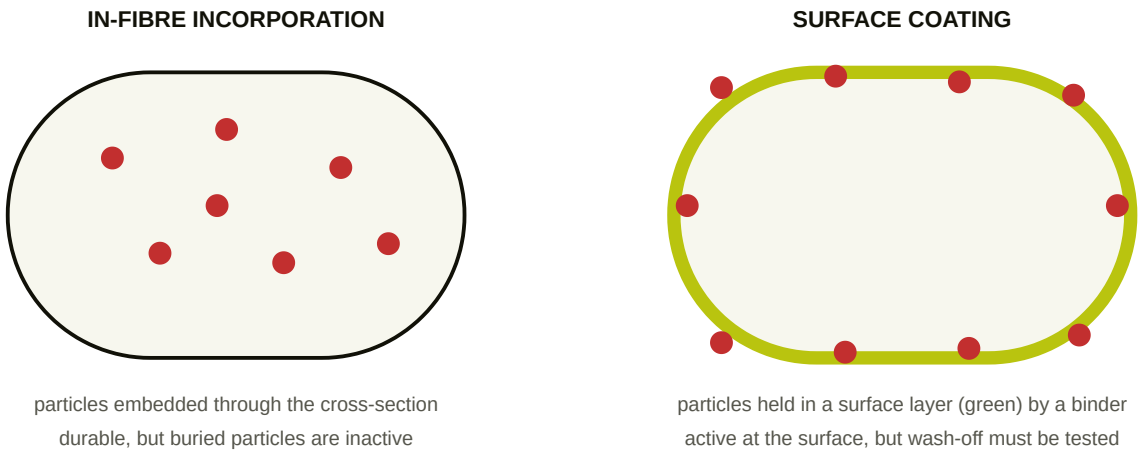


Figure 1 — The two main ways nanoparticles meet a fibre: embedded throughout the cross-section (left) or held in a surface coating (right).

4. Worked example — surface-area-to-volume scaling

For a sphere of radius r, surface area = $4\pi r^2$ and volume = $(4/3)\pi r^3$, so:

$$SA / V = 4\pi r^2 / ((4/3)\pi r^3) = 3 / r$$

WORKED EXAMPLE — WHY SMALLER IS MORE ACTIVE

Particle of radius r = 1 µm = 1×10^{-6} m: SA/V = $3 / (1 \times 10^{-6}) = \mathbf{3 \times 10^{6}\ m^{-1}}$.

Particle of radius r = 10 nm = 1×10^{-8} m: SA/V = $3 / (1 \times 10^{-8}) = \mathbf{3 \times 10^{8}\ m^{-1}}$.

Ratio = $(3 \times 10^{8}) / (3 \times 10^{6}) = \mathbf{100}$. Reducing the radius by a factor of 100 raises the surface area per unit volume by the same factor of 100 — one hundred times more surface from which silver ions can be released, or on which photocatalysis can occur, from the same mass of material.

The same scaling explains the risks as well as the benefits: a surface that reacts a hundred times more readily with bacteria also interacts more readily with skin, lungs and aquatic organisms if the particles are released. Reactivity is not selective by itself; engineering controls where the particles go.

5. Durability and safety — an honest discussion

5.1 Durability

The weak point of most nano-finishes is not the first-day performance but the twentieth wash. Particles held only by weak physical forces are progressively removed; photocatalytic particles can also attack their own binder and the fibre under UV, shortening finish life. Honest product development therefore

reports function *after a stated number of standard wash cycles*, and prefers in-fibre incorporation, covalent bonding or durable binder systems where long life is claimed.

5.2 Safety and environment

- **Release.** Nanoparticles can be released during washing, wear and disposal. Released silver, for instance, is antimicrobial in the environment too, which is a concern for wastewater organisms; wash-water studies, not slogans, are the way to quantify this for a given product.
- **Exposure.** The main occupational risk is inhalation of dry nanopowders during manufacture, controlled by enclosed handling and respiratory protection. For the wearer, particles firmly bound in or on a textile present a very different exposure from loose powder, and the two cases must not be confused in either direction.
- **Claims.** Antimicrobial and health claims are regulated in many markets. A finish that kills bacteria on a test plate does not automatically justify a medical claim for the garment.
- **End of life.** Nano-enabled textiles enter the normal textile waste stream; designers should ask where the nanomaterial ends up and whether it complicates recycling.

The balanced position is neither “nano is dangerous” nor “nano is safe”, but: *bound, tested and honestly labelled* nano-finishes can deliver real functions; unbound, untested and over-claimed ones should not be sold.

Glossary

Nanometre — 10^{-9} m. **Nanofibre** — fibre below about 1,000 nm diameter. **Nanoparticle** — particle with all dimensions roughly 1–100 nm. **Nanocomposite** — bulk material containing a dispersed nanoscale filler. **Photocatalysis** — light-driven generation of reactive species at a semiconductor surface, e.g. TiO_2. **Sol–gel** — wet-chemical route forming a thin inorganic network coating from a liquid precursor. **Plasma treatment** — surface modification using an ionised gas. **UPF** — ultraviolet protection factor of a fabric.

Key points

• The nanoscale is roughly 1–100 nm; the key driver is surface area per unit volume, SA/V = 3/r for a sphere.

• Ag gives antimicrobial action, TiO_2 gives UV absorption and photocatalytic self-cleaning, ZnO gives UV and antimicrobial action, nanoclay gives barrier effects.

• Routes: in-fibre incorporation (durable, partly wasted), coating/finishing (surface-active, wash-off risk), plasma-assisted (dry, surface-only).

• Dispersion decides nanocomposite performance; agglomerated filler is just expensive dirt.

• Durability must be reported after washing; safety assessment follows release and exposure, not the word “nano” itself.
• Every claim needs a named material, a size, an attachment method and a test.

Practice

1. Calculate SA/V for spherical particles of radius 50 nm and of radius 5 µm. By what factor does the smaller particle exceed the larger, per unit volume?
2. A mill wants an anti-odour finish that still works after 30 washes. Compare in-fibre silver incorporation with a padded silver finish and recommend a route, with two reasons.
3. Explain the photocatalytic self-cleaning mechanism of TiO_2, including why it needs UV light and one limitation of the effect in normal indoor wear.
4. Write a short, honest label claim for a ZnO UV-protective shirt: state what may be claimed, what must be tested first, and what must not be claimed.

Answer key

1. r = 50 nm = 5×10^{-8} m: SA/V = 3 / (5×10^{-8}) = $\mathbf{6 \times 10^{7}\ m^{-1}}$. r = 5 µm = 5×10^{-6} m: SA/V = 3 / (5×10^{-6}) = $\mathbf{6 \times 10^{5}\ m^{-1}}$. Factor = ($6 \times 10^{7}$) / ($6 \times 10^{5}$) = **100**.
2. Recommend **in-fibre incorporation**: (1) particles are locked in the polymer, so washing removes far fewer of them and wear exposes fresh particle-bearing surface; (2) performance after 30 washes is therefore much more likely to meet the claim. Accept the higher cost and spinning difficulty as the trade-off, and still verify by testing after 30 standard washes.
3. UV photons with energy above the TiO_2 band gap excite electrons, leaving holes; these generate highly reactive hydroxyl and related radicals at the surface, which oxidise organic soil and odorous compounds toward CO_2 and water, while the surface also becomes highly hydrophilic so water sheets off and lifts loosened soil. Limitation: indoors there is little UV, so the catalytic cycle is weak or inactive and the “self-cleaning” effect largely disappears.
4. May be claimed, after UPF testing of the finished and washed fabric: “UPF (tested value) — helps protect covered skin from UV”, naming the test standard and wash count. Must not be claimed: any medical benefit (e.g. preventing skin cancer), protection of uncovered skin, or a permanent effect if wash testing shows the finish declining.